# Ashik Sharon M

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## Summary

AI/ML engineer with hands-on experience across the field: deep learning and computer vision, NLP and large language models, retrieval-augmented generation, and the MLOps and cloud infrastructure needed to put any of it into production. Comfortable owning a problem end to end, from a research question or a messy dataset through a trained and evaluated model to a deployed, monitored system. Author of a paper under review at Elsevier, two patents in multi-agent generative AI, and merged fixes in Meta FAISS and Google DeepMind Gemma.

## Education

**Vellore Institute of Technology (VIT)** — Vellore, India
*M.Tech — Computer Science & Engineering (AI & Machine Learning); CGPA: 9.34/10* — *Aug 2024 – Jun 2026*

**Loyola-ICAM College of Engineering & Technology (Anna University)** — Chennai, India
*B.E. — Computer Science and Engineering; CGPA: 9.16/10* — *Aug 2020 – May 2024*

## Experience

**Artificial Intelligence Engineer** — Jun 2026 – Present
*CobuildX.ai* — *Chennai, India*

- Designed and shipped **AI custom agents with MCP (Model Context Protocol) tool integration** for a production AI-powered workspace platform, giving each agent scoped, real access to GitHub repositories for code-aware reviews.
- Built a **model evaluation framework** (Precision/Recall/F$\beta$, Micro/Macro-F1) with results broken down by document quality, region, and field criticality, fixing **3** real extraction defects it surfaced.
- Worked directly with product and operations teams to align model and agent behavior with real usage, and audited the production database schema for safe multi-tenant deployment.
- **Tech Stack:** Python, FastAPI, Anthropic Claude API, MCP, React, TypeScript, PostgreSQL/Supabase.

**Graduate Technical Intern — AI Engineering** — Jul 2025 – Feb 2026
*Intel Technology India Pvt. Ltd.* — *Bengaluru, India*

- Designed and deployed a production **multi-agent LLM** ticket-automation system, iterating on prompts to cut input tokens **39.4%** and model costs **35–45%** while holding accuracy steady.
- Built a Retriever API and an enterprise **GraphRAG** platform (LlamaIndex, Neo4j, FAISS hybrid search), lifting retrieval accuracy **32%** and cutting lookup latency **45%**.
- Orchestrated an **MLOps pipeline** with Docker and Kubernetes (Rancher/CaaS), shipping zero-regression releases through automated GitHub Actions CI/CD.
- **Tech Stack:** Python, OpenAI, Azure AI Foundry, LangChain, LlamaIndex, FAISS, Docker, Kubernetes.

## Projects

**MedVision-LM** | Multimodal Medical VLM Assistant — GitHub

- Built an end-to-end VLM assistant for chest X-ray analysis, achieving **89%** Recall@5 on zero-shot cross-modal retrieval using CLIP embeddings and FAISS over 7,400+ clinical images.
- Fine-tuned **BLIP** on the IU X-Ray and ROCO datasets, improving CIDEr scores **15%** for automated report generation, with GradCAM visualization reaching **92%** alignment with expert annotations.

**Zero-Shot SAR Land Cover Classification** | *Paper under review, Elsevier* — GitHub

- Designed 10 CLIP prompt templates and a hierarchical classification strategy, improving zero-shot accuracy **3.4x** over baseline with zero labeled training data, benchmarked across 27,000 images.
- Analyzed model behavior via t-SNE embeddings and confusion matrices to separate reliable "anchor" classes from semantic "confusers."

**SummarizationBench-LLM** | Trained PyTorch Model + LLM Benchmark — GitHub

- Fine-tuned **T5-small** with PyTorch/HuggingFace and **LoRA**, cutting trainable parameters **99.5%** while keeping **93%** of BART-large's ROUGE-L accuracy at **74%** lower latency.
- Benchmarked T5, BART-large, and Gemini 1.5 Pro on ROUGE-1/2/L and BLEU, with a Streamlit dashboard for live 3-way comparison.

**PoliRAG** | Enterprise Policy & Compliance Intelligence Engine — GitHub | Live Demo

- Built a RAG system with Hybrid Search (BM25 + FAISS) and cross-encoder reranking, achieving **0.70+** Recall@10 and **0.60+** MRR.
- Kept the hallucination rate under **15%** via citation-required generation and confidence-threshold filtering, with automated PII redaction and RBAC for compliance-safe deployment.

## Technical Skills

**Programming**: Python, C++, Java, JavaScript, TypeScript, SQL
**Deep Learning & CV**: PyTorch, HuggingFace (Transformers, PEFT/LoRA), CLIP, BLIP, Computer Vision, GradCAM
**NLP & GenAI**: LLMs, Prompt Engineering, RAG, LangChain, LlamaIndex, Agentic Workflows, MCP, Anthropic Claude API, OpenAI
**Classical ML & Data**: Scikit-learn, XGBoost, Pandas, NumPy, Statistics, EDA (Matplotlib, Seaborn, Plotly)
**MLOps & Cloud**: Docker, Kubernetes, AWS, Azure AI Foundry, Google Cloud, GitHub Actions, CI/CD
**Data Systems**: PostgreSQL, MongoDB, Redis, FAISS/Vector Search, Neo4j

## Research & Recognition

- **Paper:** Zero-Shot Land Cover Classification of SAR-like Imagery using Vision-Language Models (Under Review, Elsevier).
- **Patents:** Adaptive Multi-Agent Generative AI System for Legal Automation (**92% F1-score**, LayoutLMv3); Hardware-Accelerated SAR Semantic Downlink (VLM-based, **95%+** bandwidth reduction).
- **Open Source:** Meta FAISS PR #4792, Google DeepMind Gemma PR #590.
- **Certifications:** AWS (AI Practitioner, Solutions Architect, Cloud Practitioner), Google Cloud (BigQuery, Looker/LookML, Applied ML), Kubernetes.